LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Basic concept of ECM
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Basic concept of ECM. Principle of Jamming used as ECM aid.
5. Specific Objectives	18.1.4.1 Trainee will be able to appraise the function of secondary radars in ASW operation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of secondary radar ASW function diagrams; Safety: No live equipment; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a radar system?
(b) Define the term Electronic Counter Measures (ECM).
(c) How does a secondary radar assist in Anti-Submarine Warfare?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Electronic Counter Measures • Distinction between ECCM and ECM • Role of secondary radars in ASW • Basic jamming principles • Impact of jamming on detection range	L & Ds	PPT, IB	15 min
(b)	• Types of jamming signals • Mechanism of noise jamming • Mechanism of deception jamming • Secondary radar pulse compression • Analysis of ASW radar vulnerability	L & Ds	PPT, PDF	20 min
(c)	• Evaluation of jamming effectiveness • Counter-measures against jamming • Integration of secondary radar data • Operational scenarios for ECM • Summary of ECM concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how jamming affects the secondary radar's ability to detect submarines.
(b) Describe one method used to counteract jamming in ASW operations.
(c) Differentiate between the function of primary and secondary radars in ECM contexts.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of jamming discussed in this lesson will be expanded upon in the Basic concept of ECCM lesson (LP 12).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Basic concept of ECCM. Various methods of ECCM employed by Indian Naval aircraft viz. frequency agility PRF jittering etc	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min